

Istiaqur Rahman Fuad

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SUMMARY

Computer science undergraduate working across web development and machine learning. Builds full-stack apps with Next.js, TypeScript, and tRPC, and fine-tunes transformer models for low-resource languages. Published a multilingual sentence-embedding model on Hugging Face and built features for an e-commerce platform during an internship.

TECHNICAL SKILLS

Languages: C, C++, Java, JavaScript, TypeScript, Python, SQL

Technologies & Tools: Next.js, Django, tRPC, Prisma, PostgreSQL, MySQL, Git, Docker

AI / ML: PyTorch, Hugging Face, sentence-transformers, LoRA, LangChain, Qdrant (vector database)

EXPERIENCE

Advanced Machine Intelligence Research (AMIR) Lab

Feb 2026 – Present

Research Assistant Intern

Remote

- Curated and published **bangla-english-banglish-pairs**, a 2.4M-pair trilingual contrastive training dataset (Bangla / Banglish / English) on Hugging Face for low-resource NLP research.
- Trained and open-sourced **TriBne-e5** on this dataset, a Bangla/English/Banglish sentence-embedding model published on Hugging Face to improve information retrieval over code-mixed Bangladeshi text.

Devcore Technologies

Feb 2025 – Apr 2025

Next.js Developer Intern

Remote

- Built a modular admin dashboard with end-to-end type-safe APIs (Next.js, tRPC, Prisma, TypeScript) for an e-commerce platform managing a 1,000+ product catalog, delivering full product and order CRUD as part of a small engineering team.
- Developed responsive, server-rendered storefront pages with Next.js App Router, improving cross-device usability and perceived load performance.

PROJECTS

TriBne-e5-small | [Demo](#) | [GitHub](#)

- Tools used:** Python, PyTorch, sentence-transformers, Hugging Face, LoRA
- Built a sentence-embedding model for Bangla, English, and Banglish (romanized Bengali) text, matching spelling variants across scripts where standard multilingual encoders fall short.
- LoRA fine-tuned multilingual-e5-small (118M params) with contrastive loss on 2.4M pairs; achieved 0.890 MRR@10 on Banglish and 0.981 on cross-script retrieval (+0.27 / +0.48 vs. base multilingual-e5-small).

PetSoft | [Live](#) | [GitHub](#)

- Tools used:** Next.js, Prisma, TailwindCSS, Stripe, React Hook Form
- Developed a subscription-based pet daycare management system with Next.js and Prisma, featuring secure authentication, Stripe-powered subscription billing, and a responsive interface built with TailwindCSS.

EDUCATION

University of Rajshahi

Jan 2022 – Present

B.Sc. in Computer Science and Engineering

CGPA: 3.58/4.00

- Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Computer Architecture, Software Engineering, Artificial Intelligence, Machine Learning, Discrete Mathematics

COURSES AND CERTIFICATIONS

- Machine Learning Specialization – Stanford University | DeepLearning.AI (Jul 2023). [certificate](#)
- Deep Learning Specialization – Stanford University | DeepLearning.AI (Oct 2023). [certificate](#)

ACHIEVEMENTS AND ACTIVITIES

- Award of Merit** – International Blockchain Olympiad (Oct 2024). [certificate](#)
- SDG-9 Silver Medal** – International Blockchain Olympiad (Oct 2022). [certificate](#)